

AVA2

Geometry and Statistics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A landscaper who has worked for the county for eight years enlarges a square garden bed so each side is 3 times as long. How many times as much mulch area does the new bed need?

- A. 9 times
- B. 3 times
- C. 27 times
- D. 6 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Two similar cubes have volumes of 8 cm^3 and 216 cm^3 . What is the ratio of their side lengths, small to large?

- A. 3 : 1
- B. 1 : 9
- C. 1 : 3
- D. 1 : 27

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 13-foot ladder bought during a hardware store's spring sale leans against a wall with its base 5 feet from the wall. How high up the wall does it reach?

- A. About 13.9 feet
- B. 12 feet
- C. 8 feet
- D. 18 feet

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A right triangle has legs of 9 and 12. How long is the hypotenuse?

- A. 225
- B. About 7.9
- C. 15
- D. 21

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A pizzeria that delivers within 5 miles sells a 14-inch pizza, measured across its diameter. What is the area of the top of the pizza?

- A. $7\pi \approx 22$ square inches
- B. $49\pi \approx 154$ square inches
- C. $196\pi \approx 616$ square inches
- D. $14\pi \approx 44$ inches

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fish tank from a pet store with a 30-day return policy measures 30 inches long, 12 inches wide, and 16 inches tall. How many cubic inches of water fill it?

- A. 2,064 square inches
- B. 360 square inches
- C. 58 inches
- D. 5,760 cubic inches

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A roof truss designed by a firm that won a regional award is a triangle 24 feet wide. Two of its angles measure 35° and 65° . What is the third angle?

- A. 260°
- B. 100°
- C. 80°
- D. 90° , since roof trusses always have a right angle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A small firm that moved to a hybrid schedule last year has six employees with salaries (in thousands) of 40, 42, 45, 48, 50, and 205. Which measure best describes a typical salary, and what is it?

- A. The median, \$46,500
- B. The mean, about \$71,700
- C. The median, \$45,000
- D. The mean, because it uses every value and is always the best measure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Five quiz scores have a mean of 82. Four of them are 78, 85, 80, and 90. What is the fifth score?

- A.** 83.25
- B.** It cannot be found without the median.
- C.** 82
- D.** 77

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

At a school carnival that raised \$2,000 last year, a game uses a spinner with 4 equal sections, one of them red, and a fair six-sided die. What is the probability of landing on red and rolling a 6?

- A.** 1/6
- B.** 1/24
- C.** 5/12
- D.** 2/10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Events A and B are independent. $P(A) = 0.5$ and $P(A \text{ and } B) = 0.2$. What is $P(B)$?

- A.** 0.4
- B.** 0.3
- C.** 0.7
- D.** 0.1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A basketball player who practices at a gym downtown makes 80% of her free throws. Assuming shots are independent, what is the probability she misses at least one of her next two?

- A.** 0.64
- B.** 0.40
- C.** 0.04
- D.** 0.36

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A landscaper ~~who has worked for the county for eight years~~ enlarges a square garden bed so each side is 3 times as long. How many times as much mulch area does the new bed need?

Target word: area multiplier **Decides:** Area scales by k^2 with $k = 3$.

A	Correct	$3^2 = 9$.
B	Dropped element	Drops the square; that is the length multiplier.
C	Adjacent category	Uses the volume rule, k^3 .
D	Confusion set	Doubles the factor instead of squaring it.

2. Answer C Concept 1 • transfer

Two similar cubes have volumes of 8 cm^3 and 216 cm^3 . What is the ratio of their side lengths, small to large?

Target word: reverse: volume ratio to length ratio **Decides:** $216/8 = 27$; cube root is 3.

A	Inverted operation	Reverses small and large.
B	Adjacent category	Uses the area relationship.
C	Correct	Length ratio is the cube root of the volume ratio.
D	Dropped element	Leaves the ratio as a volume ratio.

3. Answer B Concept 2

A 13-foot ladder ~~bought during a hardware store's spring sale~~ leans against a wall with its base 5 feet from the wall. How high up the wall does it reach?

Target word: missing leg **Decides:** The ladder is the hypotenuse: $\sqrt{13^2 - 5^2}$.

A	Inverted operation	Adds the squares, treating the ladder as a leg.
B	Correct	$\sqrt{169 - 25} = \sqrt{144} = 12$.
C	Dropped element	Subtracts lengths without squaring.
D	Confusion set	Adds the two lengths.

4. Answer C Concept 2 • transfer

A right triangle has legs of 9 and 12. How long is the hypotenuse?

Target word: hypotenuse **Decides:** $\sqrt{81 + 144}$.

A	Dropped element	Forgets the square root.
B	Inverted operation	Subtracts the squares: $\sqrt{63}$.
C	Correct	$\sqrt{225} = 15$.
D	Confusion set	Adds the legs.

5. Answer B Concept 3

A pizzeria ~~that delivers within 5 miles~~ sells a 14-inch pizza, measured across its diameter. What is the area of the top of the pizza?

Target word: circle area **Decides:** Radius = 7 in.

A	Dropped element	Forgets to square the radius.
B	Correct	$\pi \times 7^2 = 49\pi$.
C	Wrong stem number	Uses the diameter as the radius.
D	Confusion set	That is the circumference.

6. Answer D Concept 4

A fish tank ~~from a pet store with a 30-day return policy~~ measures 30 inches long, 12 inches wide, and 16 inches tall. How many cubic inches of water fill it?

Target word: volume **Decides:** Fill = $l \times w \times h$.

A	Adjacent category	That is surface area, $2(360 + 480 + 192)$.
B	Dropped element	Base area only; drops the height.
C	Inverted operation	Adds the dimensions instead of multiplying.
D	Correct	$30 \times 12 \times 16 = 5,760$.

7. Answer C Concept 5

A roof truss ~~designed by a firm that won a regional award~~ is a triangle ~~24 feet wide~~. Two of its angles measure 35° and 65° . What is the third angle?

Target word: third angle **Decides:** Triangle angles sum to 180° .

A	Confusion set	Uses 360° , the sum around a point.
B	Adjacent category	The sum of the two given angles (the exterior angle), not the third angle.
C	Correct	$180 - 35 - 65 = 80$.
D	Absolute language	Nothing in the stem makes the truss a right triangle.

8. Answer A Concept 6

A small firm ~~that moved to a hybrid schedule last year~~ has six employees with salaries (in thousands) of 40, 42, 45, 48, 50, and 205. Which measure best describes a typical salary, and what is it?

Target word: best center **Decides:** 205 is an outlier; the median resists it.

A	Correct	Middle two values (45, 48) average to 46.5.
B	Adjacent category	$430 / 6$ is pulled up by the outlier.
C	Dropped element	Takes one middle value; with six values, average the middle two.
D	Absolute language	Using every value is why the mean is distorted here.

9. Answer D Concept 6 · transfer

Five quiz scores have a mean of 82. Four of them are 78, 85, 80, and 90. What is the fifth score?

Target word: solve for the missing value **Decides:** Total = $82 \times 5 = 410$.

A	Dropped element	Mean of the four known scores.
B	Adjacent category	The mean and count give the total; the median is not needed.
C	Wrong stem number	The mean, not the missing score.
D	Correct	$410 - 333 = 77$.

10. Answer B Concept 7

At a school carnival that raised \$2,000 last year, a game uses a spinner with 4 equal sections, one of them red, and a fair six-sided die. What is the probability of landing on red and rolling a 6?

Target word: P(A and B) **Decides:** Independent events multiply.

A	Dropped element	Drops the spinner.
B	Correct	$1/4 \times 1/6 = 1/24$.
C	Inverted operation	Adds $1/4 + 1/6$.
D	Confusion set	Pools favorable and total outcomes across two devices.

11. Answer A Concept 7 · transfer

Events A and B are independent. $P(A) = 0.5$ and $P(A \text{ and } B) = 0.2$. What is $P(B)$?

Target word: solve for $P(B)$ **Decides:** $0.5 \times P(B) = 0.2$.

A	Correct	$0.2 / 0.5 = 0.4$.
B	Confusion set	Subtracts the probabilities.
C	Adjacent category	Adds them, an “or” habit.
D	Inverted operation	Multiplies instead of dividing.

12. Answer D Concept 8

A basketball player who practices at a gym downtown makes 80% of her free throws. Assuming shots are independent, what is the probability she misses at least one of her next two?

Target word: P(at least one miss) **Decides:** $1 - P(\text{makes both}) = 1 - 0.8^2$.

A	Inverted operation	Probability of making both.
B	Dropped element	Adds $0.2 + 0.2$ and drops the overlap.
C	Adjacent category	Probability of missing both, not at least one.
D	Correct	$1 - 0.64 = 0.36$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Scale factor on area and volume	If lengths scale by k , area scales by k^2 and volume by k^3 .	Scaling area or volume by k .	"Each side is __ times as long."
2	Pythagorean theorem	In a right triangle, $a^2 + b^2 = c^2$, where c is the side opposite the right angle.	Adding or subtracting the sides without squaring, or treating a leg as the hypotenuse.	A ladder, ramp, or diagonal with two known sides.
3	Circle area vs. circumference	Area = πr^2 ; circumference = $2\pi r = \pi d$; use the radius, half the diameter.	Plugging the diameter in as r , or giving circumference for area.	A pizza, table, or pool sized by its diameter.
4	Volume vs. surface area	Volume (fill) of a rectangular prism = $l \times w \times h$; surface area (cover) = $2(lw + lh + wh)$.	Answering a fill question with surface area.	"Fill," "hold" vs. "cover," "wrap," "paint."
5	Angle sums	Interior angles of a triangle sum to 180° ; a straight line is 180° ; around a point is 360° .	Using 360° for a triangle.	Two angles of a triangle given, find the third.
6	Mean vs. median with outliers	The median resists outliers; the mean is pulled toward them. Mean $\times n$ = total.	Reporting the mean as "typical" when one value is extreme.	A data set with one value far from the rest.
7	Independent "and" events	For independent events, $P(A \text{ and } B) = P(A) \times P(B)$.	Adding probabilities for an "and" event.	Two separate devices or trials in one question.
8	Complement rule	$P(\text{at least one}) = 1 - P(\text{none})$.	Adding the single-trial probabilities.	"At least one."

Course scope (public description): Geometry and Statistics is a performance-based assessment that evaluates a student's portfolio of work. This portfolio includes the student's responses to various prompts and an original lesson plan for each of the mathematics modules such as geometry and measurement, statistics and probability.